Physical AI in Indian Agriculture

[0:09] Perfect.

[0:10] Good afternoon, everyone.

[0:11] I know this is just after lunch, so there would be a lot of hormones in your body which will tell you to sleep.

[0:18] Try your best not to.

[0:20] And if it feels like giving up, there is coffee.

[0:24] I am Shailendra.

[0:25] I am founder and CEO of a company called Pathal.

[0:28] The company is about eight years old now.

[0:31] the company started, basis of very interesting insights that we accidentally discovered.

[0:40] We discovered that India has more land than almost 150 countries combined.

[0:47] We have all kinds of climate.

[0:49] We have so many people, at least we had then, so many people.

[0:54] So technically, India should have been a food superpower.

[0:57] But then we also realized India is not one because Farmers are people who own land, not necessarily people who know how to grow crops well.

[1:09] So that's what we wanted to solve.

[1:11] The company began its journey in 2018.

[1:13] First three, four years were primarily research and development.

[1:17] And 2022 was when we launched commercially.

[1:23] My intention today through this deck is to walk you through that journey and also the ecosystem zooming in into India, zooming out of India.

[1:33] And I'm aware that there would be folks who are very well aware of agriculture here.

[1:37] There would also be folks who may not have a lot of exposure to agriculture.

[1:42] So I'll try and keep the conversation in a way that it makes sense to everybody.

[1:47] So I'll keep zooming in on very complex things and also keep zooming out on some things which are very, very intuitive to understand.

[1:58] And of course, today's conversation is about AI.

[2:01] AI has taken over the world.

[2:03] Everything is AI.

[2:04] Every company is an AI company.

[2:07] I just heard somebody asking someone, how are you using AI?

[2:10] And I think personally, the way I look at AI is it is just asking other people, how are you using Google and Excel sheet?

[2:21] AI is everywhere.

[2:22] It is the most dominating fact and it is It is one technology that is going to be far, far bigger than what digitization did, what industrialization did.

[2:31] As humans, as a civilization, we are extremely inefficient in the way that we do things, and it's by nature.

[2:39] And I look at AI primarily as the layer which is going to optimize for a lot of things which would look very intuitive 10, 20 years down the line.

[2:51] But the state of AI currently is that it's on the screen inside computers.

[2:58] And on the screen inside computers, it has started doing a lot of very great things.

[3:05] It has started creating text that humans cannot identify, whether it was created by AI or not.

[3:11] It can create pictures.

[3:13] It can create a bunch of tasks put together.

[3:18] But all of it is happening inside a computer, inside a cell phone.

[3:21] It's still a digital world.

[3:28] The next frontier of AI is going to be the physical world.

[3:31] So AI will touch living matter.

[3:35] It will touch moving cars.

[3:37] It will touch moving robots.

[3:38] It will touch farms.

[3:39] It will touch airplanes.

[3:41] It is going to touch everything.

[3:42] But that is a journey that's going to unfold in the next 10, 20, 30 years.

[3:49] And what is going to be interesting is agriculture because agriculture is the hardest frontier of AI because two farms adjacent to each other are not same, two land parcels adjacent to each other are not same, two crops grown at the same time adjacent to two farms are not same.

[4:08] And there is a very interesting AI problem, it's called cold start.

[4:11] So you train something on, models on something very, very specific and Every time you change context, you have to retrain again.

[4:19] This was a very big problem in robotics for agriculture, for example.

[4:25] But the leap that AI has made, the LLMs have made, allows you to not have a cold start problem the moment you switch context.

[4:33] So what has fundamentally changed?

[4:35] And again, there are certain other things that I'm going to talk about which are repackaging of things.

[4:41] So precision agriculture was something that was in 2010, which Wanted to do the same things, then came robotics, then now we call it physical AI.

[4:53] Most of this is rebranding, but also there has been genuine technological innovation, where some things that were not possible, but people have wanted to do them for 2030 years, are now possible.

[5:03] So now I also genuinely feel that AI will take the leapfrogging from screen to the physical world.

[5:15] But I said some context on why agriculture is something that you should care about.

[5:19] I mean, there are so many businesses with very handsome margins.

[5:23] And AI takes.

[5:25] Agriculture generally, any diffusion of technology in agriculture takes a lot of time.

[5:30] The margins work in a certain way.

[5:33] But the interesting way to look at agriculture is what if you don't have food?

[5:37] Like the moment you see what if you don't have food, you suddenly realize how expensive it is to not have the ability to grow food in the context in the environment to the quantity that it is needed.

[5:51] If you don't have food, you have a civil war.

[5:53] That's the simplest explanation I can give you.

[5:54] The last example was Sri Lanka, where somebody went and told the government that stop all fertilizers and pesticides, and then there was no food.

[6:04] But between 1960, 1950s, 60s, the human population in Italy was about 2.6 billion.

[6:11] It is about 8.2 billion right now.

[6:14] It is going to peak at around 10.3 billion.

[6:18] But there is no linearity or intuitive linearity between human population and consumption of food.

[6:27] Because generally, the world has become a richer place.

[6:31] I think the current estimated global GDP per capita is about $12,000.

[6:35] By 2050, it is supposed to be around $19,000.

[6:38] So when people become rich, they eat more nutritious food.

[6:42] They eat more food as well.

[6:44] So we have to grow more food and we have to grow more quality food.

[6:49] But what has also changed is the soil.

[6:53] We have abused the soil.

[6:54] You would have heard it again and again.

[6:56] But this is a very, very important concept to repeat.

[6:59] We have abused the soil.

[7:02] So the soil is less fertile than it was.

[7:06] The land doesn't grow.

[7:08] We have about 1.4 billion hectares of farmland in the world.

[7:12] What has happened between 1960s to 2026 is the land available to grow food per person has come down by almost 80%.

[7:23] So now we have to grow food for every person with 80% land, land that it was available in, fertile soil.

[7:33] and in climate which nobody can predict.

[7:38] That's the headway.

[7:44] What is also interesting is why one should care about India.

[7:47] So if you take a list of all the countries where there is land to grow food and you put them in descending order, you will realize that India has more land than 150 countries combined.

[8:01] It's a very big number because it also is relevant from a lot of points of view.

[8:05] So we can grow a lot of food for the world.

[8:07] That's the potential that we have.

[8:11] We have a very large agricultural market, so a lot of good businesses can be built.

[8:17] Agriculture in India is also concentrated in certain ways, which is again very counterintuitive, which I'm going to talk about.

[8:25] But India is a very fascinating example or representation of the globe.

[8:29] Because India also represents 80% of the world's land holding, so we argue again and again and we say that the land holdings are small, but the truth is 80% of the world is small, small land holding, 2015, 20% of the world is larger land holding, and that's an aberration, that's not the truth, but because technologies come from West to East.

[8:54] They are always built for large farms and that's a tricky space because we always evaluate in a way that we assume that small holdings are a problem, they actually are not.

[9:07] Now, just a refresher for everybody who's not spent a lot of time in agri.

[9:13] Agriculture works like this.

[9:15] Every crop, any part of the world, growing anything, these are the five steps any crop follows.

[9:22] You have inputs.

[9:24] So the moment the farmer decides that I'm going to grow a crop, he takes seeds or he will take a plant, he will put it in the soil, he will use water, fertilizer, pesticide in one form or another and he will start growing the crop.

[9:38] Growing the crop is also the time where the crop spends most of its time and the most miracles can happen.

[9:44] So let's say a crop, for example, let's say you eat apples, apples grow for six months on the farm.

[9:51] And they reach you in the next 10, 20, 30 days or whatever, so the maximum amount of time a crop spends is on the farm, and then of course you harvest it, you put it in cold chain for perishables, you put it in storage silos for non-perishables like wheat, rice, paddy, and then it goes to the market and to your plate.

[10:10] This is the same framework that applies to any crop grown anywhere in the world, whether on small farms or big farms, doesn't matter.

[10:21] Now, technology is in every part of this value chain.

[10:29] We have figured out a way of building better seeds because we need to.

[10:34] We used to have native seeds, which nobody knew when they will grow and when they will not grow, so we had to figure out a way of growing food continuously, so we had better seeds.

[10:43] We figured out That soil delivered a lot more to the crops, so nutrition in this and that, and we wanted to understand whether soil has enough nutrition for the crops, so we had soil tests.

[10:54] Then biologicals, a lot of technology on the input side, some chemical technologies, some digital technologies, which we are also a part of.

[11:03] And then growing, now there are sensors and satellites, which fundamentally watch and want right now.

[11:12] Why we need them?

[11:12] Because 80% of the world is still trying to figure out what is happening on their farms.

[11:17] They don't know.

[11:18] So there is need for technologies or systems or ways in which a farmer who is not very, very well versed with science knows what is happening with his crop.

[11:29] And then, of course, cold chain is there where you can decide, this is the variety, the age, at what temperature you should keep it so that it doesn't degrade.

[11:40] And then platforms are there, you have Swiggy, Zappos, the matters of the world.

[11:43] That's the downstream is the easiest to understand.

[11:46] And then it is, of course, on the plate.

[11:48] This is where you are in the world, whether you are in India or in Netherlands.

[11:53] There is penetration of these technologies in India, in Netherlands, in UK, in US.

[12:00] We pioneered sensors on the farm here in India.

[12:03] We have placed it.

[12:06] At a reasonable scale now, we cover almost 10% of all crop area for grapes in India, almost 6-7% of crop area for pomegranates in India.

[12:15] So these technologies have helped stitch the chain, build visibility into the chain, make the chain more efficient.

[12:28] But real physical AI, why is it needed in the first place?

[12:33] So India and the world is going through a very interesting transition.

[12:36] India is becoming a richer country.

[12:38] As we see today, I think Indians, it has GDP per capita is about $3,500.

[12:43] And this is a very, very interesting point.

[12:47] Whenever an economy crosses $5,000 in GDP per capita, there is a permanent migration of farm labor.

[12:57] Farm labor is what enables agriculture.

[13:01] You have to harvest.

[13:03] You have to sow, you have to irrigate, you have to fertigate, and you need a lot of labor, especially, let's say, in context of horticulture, which is nothing but fruits, vegetables, plantation crops.

[13:15] You need 9 to 10 times more labor.

[13:21] Labor is going away.

[13:22] So the moment you see India reaching 5,000 GDP per capita, you can fairly assume that if we don't solve the labor issue, which is Growing crops in situations where labor availability is a big problem.

[13:38] There is going to be a food issue, there is going to be a food shortage, there is going to be a food inflation.

[13:45] And that's why this problem needs solving.

[13:47] So how do you still continue to grow high quality crops in difficult climate with bad soil and labor not available on the farm?

[13:58] And as we become more richer, we are going to consume more fresh fruits and vegetables and things that are nutritious.

[14:04] We are going to put things in our body that are going to make us more healthier, not just fill our stomachs.

[14:11] That's why you need physical AI.

[14:12] So it's not a push on technology.

[14:15] It's a desperately needed push on problem solving, which is five years, seven years down the line.

[14:26] And what does physically I do?

[14:27] So the idea is this.

[14:28] Any farmer across the world growing any crop, again, has five things to do.

[14:34] He irrigates his crops, which is he decides whether the crop needs water or not, when does it need water, how much water it needs, and put that water.

[14:45] Then feed that crop through nutrition, which is fertilizers.

[14:49] So when does the crop actually feel hungry, how much food does it need?

[14:55] How do I deliver that food that it actually is consumed by the plant, protect it from disease issues, protect...

[15:00] Protect it from pest issues, protect it from bad climate.

[15:04] This is what every farmer does.

[15:06] Uh, if the farmer is very good, they do it very well.

[15:09] Very less number of farmers actually do that.

[15:13] A lot of farmers don't know how to make a decision about precise irrigation.

[15:18] Um, and so that's where, uh, you need that decision-making as well as you need, uh, execution of those decisions on the farms.

[15:26] And agriculture which has thrived on availability of cheap labor is going to be under severe stress.

[15:33] And the fascinating thing is, the farm is the most complex place, um, and the most hardest place to make physical AI.

[15:41] Because if you look at robots in a factory, it's a controlled environment.

[15:46] If you look at cars on the road, it is still is a controlled environment that the car has to move between two lines.

[15:54] On a farm, nobody knows what is happening at which part of the farm.

[15:59] Two farms adjacent to each other could also really, really struggle to have same output, very, very different conditions.

[16:07] So getting something to work here is a very hard thing to do.

[16:11] But the good news is, people have been trying, um, their hands at it for a few years now, at least 10, 15 years since I've been tracking this industry.

[16:22] Um, of course, the latest, uh, LLMs have really enabled some of the problems which were far stuck.

[16:28] So for example, Carbon Robotics is a company, uh, that does make very large, uh, weeders, which can see where is weed on the farm, and they can kill it through laser.

[16:39] Uh, brilliant, beautiful machine because a lot of VV side, which is chemical to, to be sprayed on the weeds is used, uh, which is consumed by humans because it stays in the plant and it stays in the fertility as well.

[16:53] Then there is Eco Robotics, um, again, plant by plant.

[16:57] So these machines are seeing the plant, they are making an assessment of what is needed, and then they are acting on that problem statement.

[17:06] This is, uh, the machine that I was talking about.

[17:09] It's a beautiful machine, but look at how big this is, and make your assumption on whether it can work in an Indian farm.

[17:17] And that's why you need India-oriented solutions, which will also work for 150 other countries.

[17:23] And now, in the last 10 years, in front of my own eyes, I have seen the cost of sensors coming down, the cost of, uh, satellite imagery coming down.

[17:32] So a lot of, uh, good things have happened, uh, in terms of seeing.

[17:37] So we can see better, we can see for a lesser cost now.

[17:40] But the value now thereby has shifted on modeling.

[17:43] Uh, modeling is, after seeing, how do I make an assessment as a model of what is right or what is wrong, and what do I do to act on it or solve that problem.

[17:53] And it becomes a loop.

[17:54] So you go to more farms.

[17:55] That is what I've seen happening with us.

[17:58] You go to more and more farms, you get more data, and every season your models are becoming better, and you go to enough farms that your models become very, very differentiated and better.

[18:09] That's where value is going to lie.

[18:11] And why horticulture?

[18:13] We, as an organization in 2018, picked up horticulture.

[18:16] It is a subsegment within Indian agriculture.

[18:19] So Indian agriculture is about 150 million hectares of farmland.

[18:23] Think of it like 3,000 Singapores.

[18:25] Uh, Indian horticulture, which is a subsegment there, is about 30 million hectares.

[18:30] But funnily enough, that 30 million hectares produces almost 40, 45 percent of dollar value from agriculture output point of view.

[18:38] And because the farms are complex, every crop is different, uh, every farm operation is different, uh, it's a difficult place to build these kind of technologies.

[18:49] But it is also a place where building these technologies can have outlasting business impact.

[18:54] And again, I spoke about the Lewis turning point, but that's where India is headed.

[19:00] So India needs ability to see the farms, to analyze what is happening on them, and act on those decisions or circumstances and situations, uh, in a labor scarce situation for lesser cost than what laborers are getting paid for right now.

[19:15] And that's the opportunity.

[19:16] This is who we are.

[19:18] This is what we do.

[19:19] Um, we were a precision agriculture company, but now people understand deep tech.

[19:24] So I'm still a little bit confused on how should I position, whether it is a deep tech company or a physical AI company.

[19:32] And that's the conundrum of our times.

[19:34] But essentially, what we do is we build sensors that are reliable on the farms.

[19:39] They can survive farm conditions.

[19:41] So they can be at 55 degree centigrade temperature, minus 20 degree centigrade temperature, rainfall of 100 mm in a day doesn't matter.

[19:50] What we built for is reliability of these sensing systems on the farm.

[19:54] And then the next thing that we built for is how do you consume all of this data and help a farmer who doesn't understand science.

[20:02] And that's where AI comes into picture.

[20:04] Tell him in very, very simple way that your crop requires water, and this is the exact amount of water that is needed.

[20:12] That your crop is going to get sick.

[20:14] You don't see it right now, but it is going to get sick.

[20:17] If you treat it right now, then it's going to be a 100 rupee game.

[20:21] But if you treat it 20 days later, it's going to be 2,000 rupees per acre.

[20:25] That was the next thing that we built.

[20:28] And now, with a lot of labor pressure, especially horticulture, because it is always surrounded, or it is always clustered in, uh, areas which are very close to, uh, big cities.

[20:38] So the labor there is competing with SMEs, with, uh, with, uh, gig economy jobs, with, with construction jobs, which pay far better.

[20:46] And thereby, there is a deep pressure within Indian horticulture of figuring out labor कहां से मिलेगा.

[20:53] Labor is what physical AI potentially is going to solve.

[20:57] The technology is very simple.

[20:59] See the farm through sensors.

[21:01] All of these decisions that the farmers have to make, they all depend on a variable.

[21:06] Collect all of those variables.

[21:09] Build model which understand the interplay of these variables.

[21:13] So they understand the relationship of these variables to derive whether the crop needs water or not.

[21:20] We worked on that quite a lot.

[21:22] Uh, our earlier assumption was sensors as a concept will be a differentiating factor.

[21:27] But later we realized, no, ideally the cost of hardware will continue to come down.

[21:33] And so we then produced this intelligence and shared this intelligence with farmers, uh, on their mobile phones, on their WhatsApp, wherever they are, but in very simple manner that they can consume.

[21:46] Somebody who has, who doesn't even know how to read can just hear.

[21:51] And then the next thing is, how do we build machines and technologies which are relevant for India, can survive Indian margin pressure, can build an Indian business, but execute all of these decisions without the farmer having to have, uh, the worry of whether there is labor to execute that task or not.

[22:11] The first thing that we have done is execute irrigation and execute fertigation, which is putting water without any labor and put in labor.

[22:20] Um, we have deployed about 13, 14,000 IoT devices.

[22:23] This would also be the biggest IoT deployment, at least in this part of the world for farm.

[22:28] Uh, these devices collect a lot of data about, uh, canopies temperature, the leaf wetness on the leaves, the moisture in the soil in real time, every hour, every half an hour.

[22:39] Um, we have about 10 billion data points like that now.

[22:43] Uh, we have deployed it across 200,000 acres of farmland with individual farmers plus 120 enterprises.

[22:49] Uh, we have been collecting this data for last five, seven years.

[22:53] It has become very big datasets.

[22:55] Also has created a lot of differentiation.

[22:57] And then act is where FossilJet and FossilJet Pro come into picture.

[23:02] So they can act on the recommendation that system is generating.

[23:06] Um, this is how the machine that does irrigation and fertigation look like.

[23:10] And the result of the technology in last three, four years is that we have saved about 83 billion liters of water.

[23:17] So this is the water that, uh, ideally you would get to drink, but goes in the farms.

[23:24] Uh, 83 billion liters of water.

[23:26] If you want to imagine, imagine the Powai Lake and multiply it by 83 times.

[23:31] About 210,000 kilograms of pesticides that you could have ultimately consumed was avoided on these farms.

[23:38] Uh, 56,000 metric tons of GHG emissions.

[23:40] So think of entire Mumbai, or think of Colaba as an area, and, uh, assume that the air was clean.

[23:47] So we consumed, sucked up everything that was making the air, uh, not so clean.

[23:52] And this is a multi-country approach.

[23:54] I mean, the fact is, what you build in India is very relevant for a lot of parts of the world.

[24:01] Um, you can hear directly from what the technology does to a very average farmer who probably doesn't even, uh, even know how to speak anything beyond his local language, but has to complain this now.

[24:24] મારું નામ કિશોરકુમાર તુલસીદાસ લખ અને ગામ પાંચિયા.

[24:27] શરૂઆતમાં અમે જ્યારે ખેતી કરતાં ત્યારે નોર્મલી સાદા પાકની જ વાવેતર કરતાં.

[24:32] અને 2009 માં આવ્યા પછી અમે દાડમની ખેતી ચાલુ કરી છે.

[24:35] અને દાડમની ખેતીમાં જોઈએ તો શરૂઆતની અંદર અમારે રોગ જીવાત ઓછા આવતા.

[24:40] અત્યારના સમયમાં જોઈએ અને અત્યારના ક્લાઈમેટ પ્રમાણે જોઈએ તો રોગ જીવાત અત્યારે બહુ વધારે છે.

[24:46] તો એમના પાછળ અમે રિસર્ચ કર્યું અને કારણ જોયું તો પાણી જમીનની અંદર વધારે આપવાથી અને એના લીધે પણ રોગ જીવાત વધારે આવે છે.

[24:53] તો એના માટે અમે આ ફસલ નામની જે કંપનીનું મશીન આવે છે, તો એ મશીન અમે વાડીની અંદર ઇન્સ્ટોલ કરેલું છે.

[25:00] તો આપણી પાસે જે ફસલ નામનું મશીન છે, એમની પાસે ટેકનોલોજી છે કે જેઓ આપણને ત્રણ થી ચાર દિવસ અગાઉ કહી દે છે કે આ ક્લાઈમેટ પ્રમાણે તમારા ખેતરની અંદર, તમારા પાકની અંદર આ રોગ જીવાત આવી શકે છે.

[25:12] દાડમની અંદર અમે ફસલ મશીન લગાવ્યા પછી જે ફસલ મશીનની અંદર નોટિફિકેશન આવે છે કે જે તમારે આ સ્પ્રે કરવાનું.

[25:19] તો અમે શરૂઆતમાં પહેલા જ્યારે મશીન નહોતું, ત્યારે અમે આની અંદર સ્પ્રે કરતા.

[25:23] અને હવે જે મશીન આવ્યા પછી અમે સ્પ્રે કરવાથી અમને પાંચ થી સાત સ્પ્રે ઓછા થઈ ગયા છે.

[25:29] અને જે એનો ખર્ચ છે, એ અમારે આ સ્પ્રે ઓછા કરવાથી જે મશીનનો ખર્ચ અમને સ્પ્રેના ખર્ચમાંથી જ નીકળી ગયો છે.

[25:36] દાડમ પાકની અંદર અમે જ્યારે મશીન નહોતું લગાવેલું, ત્યારે જે છોડને ખાતરની જરૂરિયાત રહેતી, એ ખાતર અમે પરંપરાગત રીતે આપતા કે જે છોડને ડ્રીપની અંદર આપતા, તો એ ખાતર નીચે ઉતરી જતું અને છોડ પૂરું ખાતર ઉપાડતું નહીં.

[25:49] પણ જ્યારે આ ફસલ મશીન લગાવ્યું, એમની અંદર વીપીડી નામનું નોટિફિકેશન આવે છે.

[25:54] નોટિફિકેશન અમે જોઈ અને છોડને જેટલી જરૂરિયાત હોય એ પ્રમાણે ખાતર આપતા કે જે પાણીને ટાઈમ ટુ ટાઈમ આપીને જેટલી જરૂરિયાત હોય એ પ્રમાણે જ આપતા કે જે ખાતર નીચે ઉતરતું બંધ થઈ ગયું અને છોડ પૂરેપૂરું ખોરાક ઉપાડતો થયો.

[26:07] અત્યારે ખેતી કરતા ખેડૂત મિત્રોને એટલું જણાવીશ કે અત્યારની આ આધુનિક પદ્ધતિમાં જે રીતના આપણે ખેતીમાં ડ્રીપ ઇરીગેશન આવશ્યક છે, એ જ રીતના વાતાવરણને અગાઉ જાણવું એના માટેનું આ ફસલ કંપની દ્વારા જે ટેકનોલોજી વિકસાવવામાં આવી છે, એ ટેકનોલોજી અપનાવવી જોઈએ.

[26:23] જેનાથી આપણે 14 દિવસ અગાઉનું આપણે એપ્રોક્ષ વાતાવરણ જાણી શકીએ છીએ,

જે તમને 80 થી 90% ચોકસાઈ પૂર્વકનું તમને વાતાવરણ અને માહિતી આપશે.

[26:32] જેનાથી આપણે ખેતીની અંદર ખર્ચ ઘટાડી શકીએ છીએ, ઉત્પાદન વધારી શકીએ છીએ અને જે ઉત્પાદન મળે છે, એ ક્વોલિટી ઉત્પાદન મળે છે.

[26:39] જેનાથી આપણને જે ઉત્પાદિત થયેલ આપણું જે છે પ્રોડક્ટ, એમાં આપણને ભાવ સારા મળે છે, જેથી ખેતીની અંદર નફો વધે છે.

[26:46] એટલે એથી આપણે આ મશીન લગાવવું જોઈએ, જેથી પાણીની અંદર મેન પિયતમાં બચત થાય છે.

[26:51] પિયતમાં બચત થતાં ખાતરમાં બચત થાય, નિંદામણમાં બચત થાય છે અને ઉત્પાદનની ગુણવત્તામાં વધારો થાય છે.

[27:01] અને સાથે સાથે આ અગાઉ આપણે વાતાવરણ જાણી શકતા હોવાથી સ્પ્રેમાં પણ એક બે સ્પ્રે ઘટે છે, જેનો સીધે સીધો આપણો ખર્ચ, મશીનનો ખર્ચ સીધે સીધો એક બે સ્પ્રે ઘટવાથી પણ નીકળી જાય છે.

[27:18] એટલે ખેડૂત મિત્રોએ ખાસ આ ટેકનોલોજી અપનાવવી જોઈએ અને આધુનિક સાથે હરણફાળ કદમથી કદમ મિલાવવો જોઈએ.

[27:27] There are about 13,000 of them right now.

[27:30] Uh, we are hopeful that by end of this financial year we should be about 20,000.

[27:35] Um, and I see the next 10 years for us, as well as the world, a journey of taking all of this screen intelligence to something that truly can, uh, work on the farms in complex situations as well.

[27:49] Uh, this is us.

[27:50] Um, we are called FossilJ.

[27:51] Um, we are a team of about 120-odd people, patented technology, working with farmers since 2022 in about 30-odd districts in India and about six countries now.

[28:02] That's all from me.

[28:04] Thank you so much.

[28:32] I can absolutely.

[28:33] Uh, thanks for the good session, Shalinder.

[28:35] Uh, I think first, if you could just let us know what the business model is.

[28:38] So how do you charge the farmers?

[28:40] Is it for the device and then incrementally for, uh, data sharing and reports and analysis with them?

[28:45] Oh, certainly.

[28:46] So we sell the hardware.

[28:47] Uh, we also attempted subscriptions, but we realized subscriptions are very, very difficult.

[28:53] Um, and farmers are technically more comfortable with, uh, CapEx than Opex.

[28:58] So right now we are on a journey of converting that, uh, uh, you know, Opex to CapEx.

[29:04] So we sell the hardware, all kinds of hardware, and then there is a, there is, there is a small percentage of subscription over the life of the hardware.

[29:14] Got it.

[29:15] Okay.

[29:15] And does the model, as you said, you focus on the horticulture phase.

[29:19] Is it just primarily, as you said, because much more value per acre is there and hence it makes commercial sense in the future?

[29:27] Or do you see this as sensor cost come down, as you said, can be used for wheat, rice, etc.

[29:33] as well?

[29:34] See, the technology can be applied on any crop because the fundamentals remain same.

[29:40] Now, there are two things.

[29:42] One is who is ready to adopt technology.

[29:44] Somebody whose yield grows by 20% and makes one and a half lakh rupees or one lakh rupees additional per acre has an inherent incentive of looking for these technologies and adopting them and scaling them.

[29:58] The challenge.

[30:00] That I see with crops like wheat, rice, etc., at least directly with the farmer.

[30:05] So, within agriculture ecosystem, there are a variety of customers.

[30:09] You have the farmer, you have the district, you have the government, and everybody has a lot of vested interest.

[30:16] So, I'm— I'll share something very interesting: the Government of India's agriculture budget is 1,22,000 crores.

[30:24] Additionally, the fertilizer subsidy budget is 1,70,000 crores.

[30:28] So the budget for fertilizer subsidy is bigger than the agri budget for an entire year.

[30:33] And so, from that point of view, the government could pay.

[30:37] But from a farmer's standpoint, what— what I still continue to struggle with is the cost of explaining the value of the technology.

[30:46] In itself, it is a deal breaker right now.

[30:48] Maybe 5 years, 7 years down the line, there would be a different business model which would enable something else.

[30:56] But the technology is applicable in all crops.

[30:59] The question is: who is ready to adopt, and where should I focus very limited resources that I have?

[31:06] Hence, horticulture person.

[31:08] Absolutely.

[31:09] Got it.

[31:09] And you quantify the impact it had on pesticide usage, water usage.

[31:13] Possible to quantify the impact it had on yield, let's say, and fertilizer usage as well?

[31:19] Certainly.

[31:19] So, um, in general we see a 15% to 300% improvement.

[31:23] And it depends on crop to crop.

[31:25] So 300% improvement, not because we are doing something miraculous, but the ways in which we grow crops is very varied.

[31:34] Let's say in vegetables, I have seen on an average about 20, 25 percent improvement.

[31:40] In fruits, I've seen anything between 10 to 20 percent improvement.

[31:45] Um, which is enough, and it recovers the cost of the technology for all of these crops within an year.

[31:52] In terms of fertilizers, we don't directly measure because we can't measure.

[31:58] For water, we can directly measure.

[32:00] For GHC emissions basis, the number of space sprays reduced, number of pump hours reduced, we can directly measure.

[32:09] There is an estimation that we— we have saved about 15% fertilizers every time, on every farm.

[32:16] But— but that's a non-audited data.

[32:18] The numbers that I'm sharing are audited numbers.

[32:23] Makes sense.

[32:24] Super.

[32:24] Another audio question I— question I see is: you, as you mentioned, you have lots of sensors spread across the country now.

[32:32] Yeah.

[32:32] And you're collecting tons of data on this.

[32:34] Can that data be used for other reasons, or be sold commercially as well?

[32:39] Well, absolutely, yes.

[32:40] And I think that's the bigger, more beautiful part of this entire, uh, play.

[32:45] What we are collecting is a lot of climate, farm, and crop data.

[32:49] Climate, farm, and crop data can be used in a variety of ways.

[32:53] The easiest— so for example, some of you might be aware that IMD has about 4,000 weather stations in the country.

[33:01] They have been installing weather stations from 1947, I believe.

[33:05] We have about 14,000 now.

[33:06] So the— and the country needs at least 200,000 stations before we can make any reasonably good weather prediction.

[33:14] The reason why we don't make very good weather prediction is because we don't have the infra.

[33:20] So that's point number one.

[33:21] Second is a lot of commercial usage of this data.

[33:24] So for example, at this point in time, I know how many farmers are having a good season for growing grapes in Nashik.

[33:32] I know whether the prices will be good or bad, because you have a representative data of whether the season allowed for a group to— a good crop to happen or not.

[33:42] And so that's very useful for buyers.

[33:45] That's very useful for supply chain players who want to proactively manage their procurement.

[33:51] Or input companies.

[33:52] So input companies decide what they are going to place on the stores in retail chains.

[33:57] Basis what happened last year.

[33:58] They lose sales because there are often times where you need very specialized high-margin product, which is not available because nobody has the real-time data.

[34:07] So there is a variety of application of this data.

[34:10] There's carbon credits, water credits data.

[34:12] So every— every farm we go to has reduced water consumption at least by 30%.

[34:16] That in itself converts to a lot of savings.

[34:19] So that's another angle that we are looking to play, and probably that could be a very large angle to play as well.

[34:25] But right now you're not commercializing it, let's say for— Not right now.

[34:28] Not right now.

[34:29] So the question was: do you have large enough dataset?

[34:32] And I think that question is now answered.

[34:34] Now we are trying to pursue some of those things.

[34:37] Okay.

[34:38] Makes sense.

[34:38] And were you able to customize the solution region to region?

[34:41] Each region in India is different, the crops are different.

[34:44] So it's a myth that you need to customize for countries and regions.

[34:49] It's— it's to say, um, that if I go get cold here in India, am I going to get cold in Australia or not?

[34:57] The fundamentals of science across the world, within the physical boundaries of biology, are same.

[35:04] A plant requires water for the same reason in India as it requires it in Kenya, or as it requires it in Europe.

[35:12] That mathematics doesn't change.

[35:14] So what changes is very minor, um, delta on what we call resistance building.

[35:20] So in certain areas, you might— you might have very robust plant varieties which are resistant— resistant to a certain problem.

[35:29] And that's a very easy thing.

[35:31] And— and see, also the fact that we did R&D for 5, 4 years before we commercially launched was because we wanted to build science-first models which will scale across other parts of the world.

[35:45] So the delta that is needed is actually very, like, small.

[35:49] It is not more than 15% of work that needs to be done.

[35:54] Makes sense.

[35:55] Another question I see here is: you have 100 models, as you said.

[35:59] How do you decide which model to use and when?

[36:02] Given that wrong decisions can impact crop season.

[36:05] I can go back.

[36:07] Um, not on my slide.

[36:08] Uh, I'm going to use a slide to paint a picture of when for what do you need a model.

[36:14] See, essentially, we are working on the growing phase of the crop, where the farmer is growing the crop.

[36:22] Where the farmer is growing the crop.

[36:25] And these are decisions that people have to take every day, no matter what.

[36:31] They have to decide whether there is requirement for water.

[36:35] It's a non-negotiable.

[36:37] It's— it's like, if you don't feed the crop, it is going to die.

[36:42] Then managing disease.

[36:43] So the crops will get sick.

[36:46] That is the nature of biology.

[36:48] So you have to protect them from disease, pest issues, climate risk issues.

[36:53] So for every crop, and every crop at variety of its crop stages are different.

[36:59] So you might see a small plant like this.

[37:02] It will require very different amount of water.

[37:06] And a tree 10 feet tall will require very different amount of water.

[37:11] So what we need to do is, number one, predict the stage of the crop without seeing the crop on the farm.

[37:19] So that's algorithm number one.

[37:21] That is common across all the crops.

[37:25] Second is irrigation requirement.

[37:27] So by crop, by stage of the crop, by different textures of soil in different locations on different altitude, you have to figure out how much water the crop needs.

[37:37] So that's the second algorithm.

[37:39] The 200 algorithms that I spoke about are disease and pest algorithms.

[37:43] So that's something which is different.

[37:45] So you might have some diseases here in India and some additional diseases in Thailand.

[37:51] Which one we keep or we build is a function of which market we are dealing with.

[37:56] And then climate risk management is fundamentally a common subject.

[38:00] I mean, it's not just India.

[38:01] In every part of the world where climate was something that farmers intuitively understand.

[38:07] Now there is a 35 degree centigrade in a place in Thailand where the temperature shouldn't go beyond 24, 25.

[38:14] So the climate risk management has also become a common theme across.

[38:18] And these are the models that we build.

[38:21] Perfect.

[38:22] I think those were the questions that we covered.

[38:24] Thank you so much.

[38:25] Thank you so much.

[38:26] Thank you, everybody, for your time.

[38:35] Do you want water?

[38:36] Do you want water?